AWS VPC Peering Project Report

Student Name: Vedant Thapliyal

Project Title: AWS VPC Peering

AWS Region: Asia Pacific (Mumbai) — ap-south-1

Project Overview

The objective of this project was to configure VPC Peering between two Amazon Virtual Private Clouds (VPCs) and test private network connectivity between their EC2 instances. The project included VPC and subnet configuration, route table setup, VPC Peering, security group configuration, and connectivity testing.

2. VPC Architecture and Configuration

2.1 Project Architecture

This project uses two Amazon Virtual Private Clouds connected through VPC Peering.

- VPC-A CIDR: 10.10.0.0/16
- VPC-B CIDR: 10.20.0.0/16
- AWS Region: Mumbai (ap-south-1)
- VPC Peering Connection ID: pcx-078e8e902dd4bfbea

2.2 Subnet Configuration

VPC	Subnet Type	CIDR Block
VPC-A	Public Subnet	10.10.1.0/24
VPC-A	Private Subnet	10.10.2.0/24
VPC-B	Private Subnet	10.20.1.0/24

2.3 EC2 Instances

Instance	Private IP Address
Bastion Host	10.10.1.215
App-A	10.10.2.222
App-B (new instance)	10.20.1.233

2.4 VPC Peering

The VPC Peering connection was created between VPC-A and VPC-B. The connection status was verified as Active.

Routes were configured to allow traffic between the two VPC CIDR blocks through the peering connection. Connectivity tests were performed and their actual results are documented in the testing section.

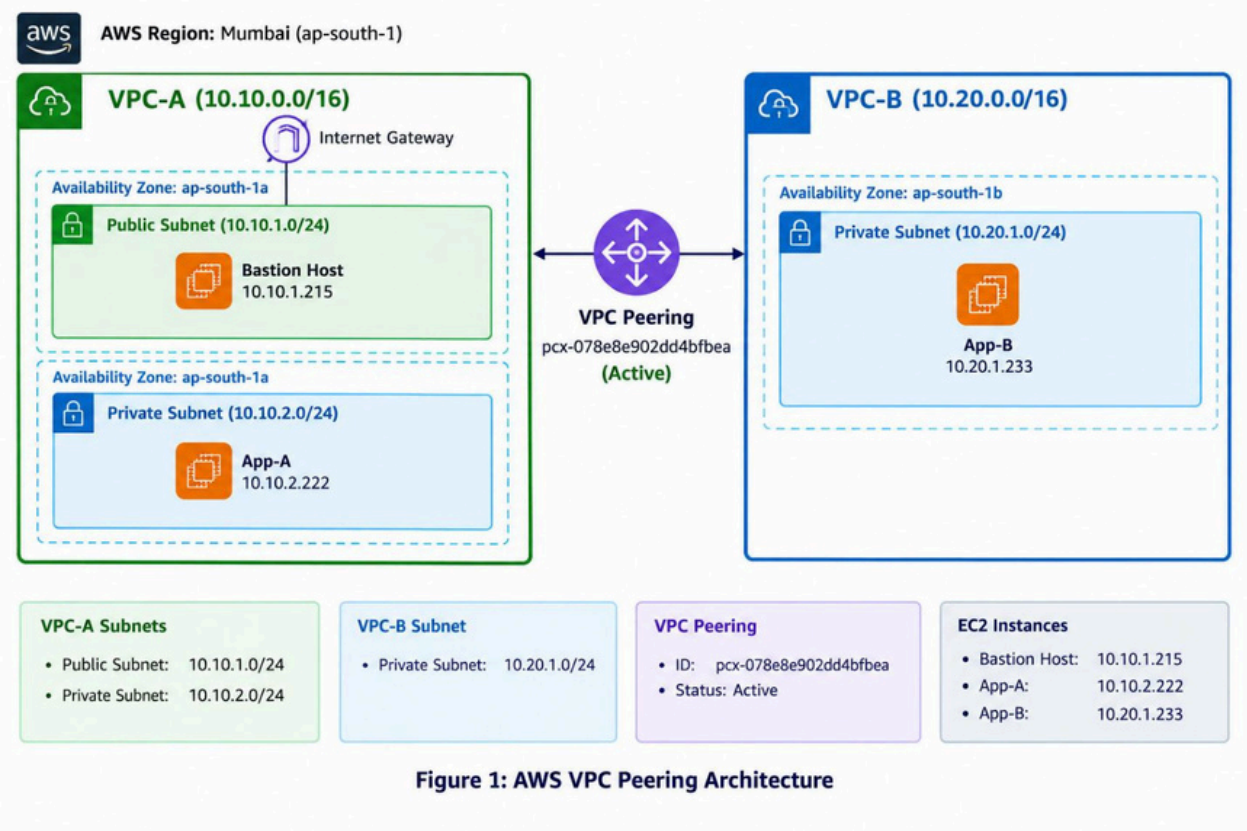


Figure 1: AWS VPC Peering Architecture

3. Route Tables and Security Groups

3.1 Route Table Configuration

Route tables determine how network traffic is directed between the VPCs. Routes to the remote VPC CIDR were configured through the VPC Peering connection.

Route Table 1: ewb-rt-a-public

- VPC: VPC-A
- Associated Subnet: Public subnet (10.10.1.0/24)
- Local Route: 10.10.0.0/16
- Internet Route: 0.0.0.0/0 through the Internet Gateway
- Peering Route: 10.20.0.0/16 through pcx-078e8e902dd4bfbea

Route Table 2: ewb-rt-a-private

- VPC: VPC-A
- Associated Subnet: Private subnet (10.10.2.0/24)
- Local Route: 10.10.0.0/16
- Peering Route: 10.20.0.0/16 through pcx-078e8e902dd4bfbea

Route Table 3: ewb-rt-b-private

- VPC: VPC-B
- Associated Subnet: Private subnet (10.20.1.0/24)
- Local Route: 10.20.0.0/16
- Peering Route: 10.10.0.0/16 through pcx-078e8e902dd4bfbea

3.2 Security Group Configuration

Security groups act as virtual firewalls for EC2 instances. Inbound and outbound rules control which network connections are allowed.

4. Connectivity Testing and Results

4.1 Objective

The objective of connectivity testing was to verify network communication between EC2 instances deployed in separate VPCs connected through VPC Peering.

4.2 Ping Testing

ICMP ping tests were performed from the Bastion Host to check connectivity with the application instances.

Test Source	Destination	Result
Test 1 Bastion Host (10.10.1.215)	App-B (10.20.1.233)	Refer to actual screenshot
Test 2 Bastion Host (10.10.1.215)	App-A (10.10.2.222)	Refer to actual screenshot

Test 1: Bastion Host to App-B

The Bastion Host was used to run a ping test toward App-B's private IP address. The command output is included as evidence.

Screenshot: Bastion_to_App-B_Ping.png (uploaded in github)

Test 2: Bastion Host to App-A

A second ping test was performed from the Bastion Host toward App-A's private IP address. The command output is included as evidence.

Screenshot: Bastion_to_App-A_Ping.png (uploaded in github)

4.3 SSH Connectivity

SSH connectivity between the application instances was not confirmed during this lab session. The SSH troubleshooting process helped identify the importance of checking routing, security group rules, network reachability, and key-based authenticatio

5. Conclusion and Key Learnings

5.1 Conclusion

This project provided practical experience in configuring Amazon VPC Peering between two Virtual Private Clouds in the AWS Mumbai Region.

The implementation involved creating and configuring VPCs, subnets, route tables, security groups, and EC2 instances. Connectivity tests were performed from the Bastion Host to the application instances, and the results were recorded using terminal screenshots.

The project also provided practical experience in identifying and troubleshooting network connectivity issues in AWS.

5.2 Key Learnings

- Understanding VPCs and CIDR blocks.
- Creating public and private subnets.
- Configuring route tables and subnet associations.
- Understanding VPC Peering connections.
- Configuring security groups for SSH and ICMP traffic.
- Testing connectivity between private IP addresses.
- Troubleshooting network connectivity and SSH access.
- Capturing screenshots and documenting AWS infrastructure.

5.3 Skills Practiced

AWS Services: Amazon VPC, EC2, VPC Peering, Route Tables, Security Groups, Internet Gateway, and AWS Systems Manager Session Manager.

Networking Concepts: CIDR, private IP addresses, routing, ICMP, SSH, and network troubleshooting.

5.4 Future Improvements

- Complete and verify direct SSH connectivity where required.
- Capture additional evidence for the remaining acceptance criteria.
- Review resource cleanup and verify that lab resources are no longer running when the project is finished.

5.5 Final Note

This project strengthened my understanding of AWS networking fundamentals